Performance Analysis of UWOC System

Using OOK and 4-QAM with Diversity Techniques

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Outline

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Introduction

- Underwater communication is essential for ocean monitoring, defense, and exploration.
- Acoustic systems provide long range but suffer from low data rate and high latency.
- RF signals face severe attenuation in water.
- **UWOC advantages:**
  - High data rate
  - Low latency
  - Large bandwidth
- **Challenges:**
  - Absorption
  - Scattering
  - Distance
  - Water conditions

Problem Statement

Objective

Design a UWOC system using OOK and 4-QAM, apply diversity techniques, and analyze BER under varying SNR, distance, and water conditions.

Objectives

- Implement OOK and 4-QAM modulation
- Model UWOC channel using attenuation
- Analyze BER under varying:
  - SNR
  - Distance
  - Water conditions
- Apply spatial and frequency diversity
- Compare performance of both modulation schemes

System Architecture

Transmitter

- Random bit generation
- OOK / 4-QAM modulation

Channel

- Beer-Lambert attenuation
- AWGN noise
- Spatial + Frequency diversity

Receiver

- Equalization
- Detection
- BER computation

UWOC Channel

- Optical propagation affected by absorption and scattering
- Depends on water type:
  - Clear
  - Coastal
  - Turbid
- Higher attenuation $\Rightarrow$ higher BER

Channel Model Equation

Beer-Lambert Law:

$$P_r = P_t e^{-cd}$$

- P_t : Transmitted power
- P_r : Received power
- c : Attenuation coefficient
- d : Distance

Observation:

- Increasing distance reduces received power
- Higher c degrades performance

OOK

- Binary intensity modulation
- $1 \rightarrow$ ON, $0 \rightarrow$ OFF
- Simple and low-cost
- Sensitive to channel degradation

4-QAM

- Maps 2 bits per symbol
- Higher spectral efficiency
- More complex than OOK
- Sensitive to noise and attenuation

Spatial Diversity

- Multiple transmission paths
- Combines signals at receiver
- Reduces fading
- Improves BER

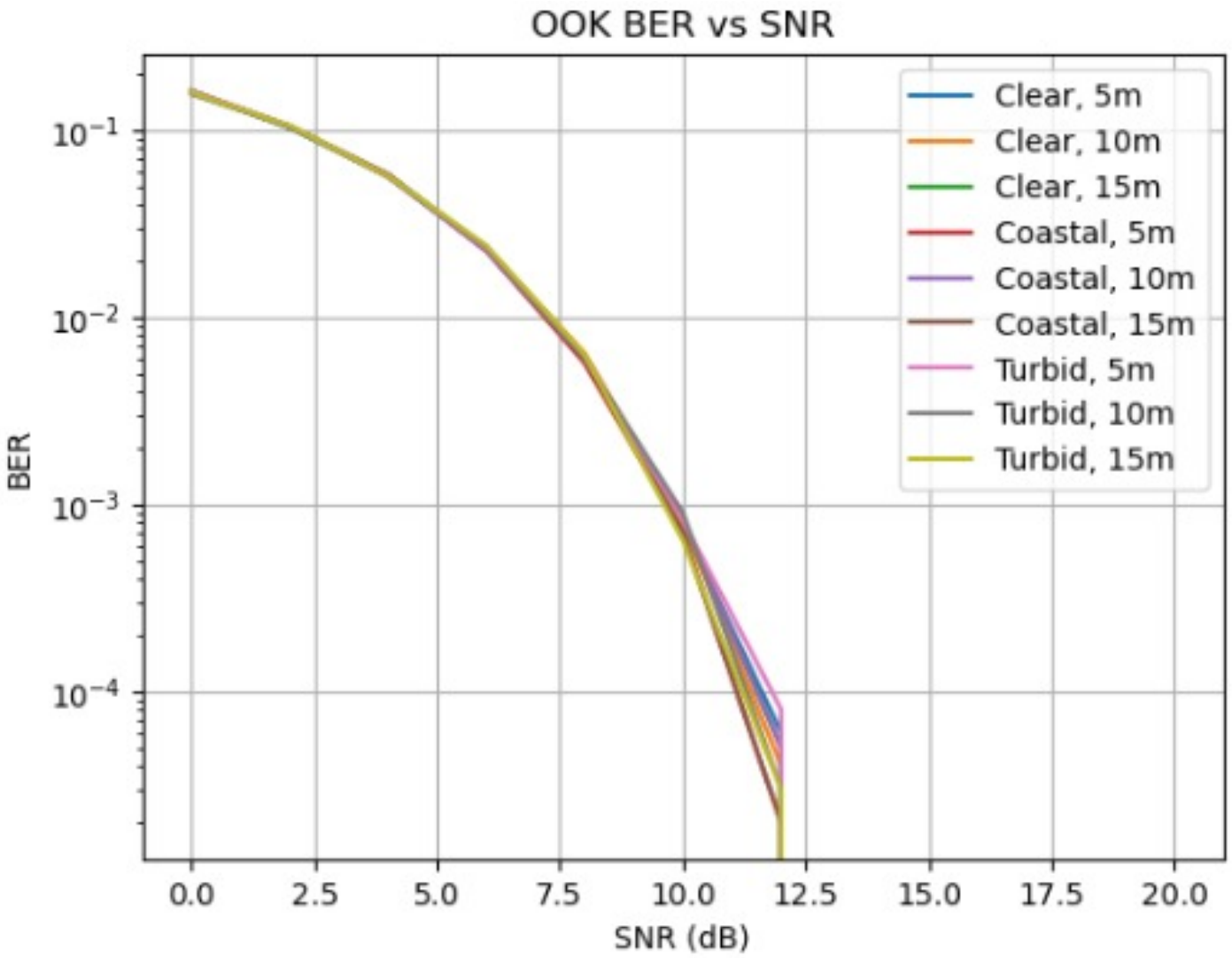
Frequency Diversity

- Same signal over multiple frequencies
- Different attenuation per branch
- Improves reliability

Expected Trends

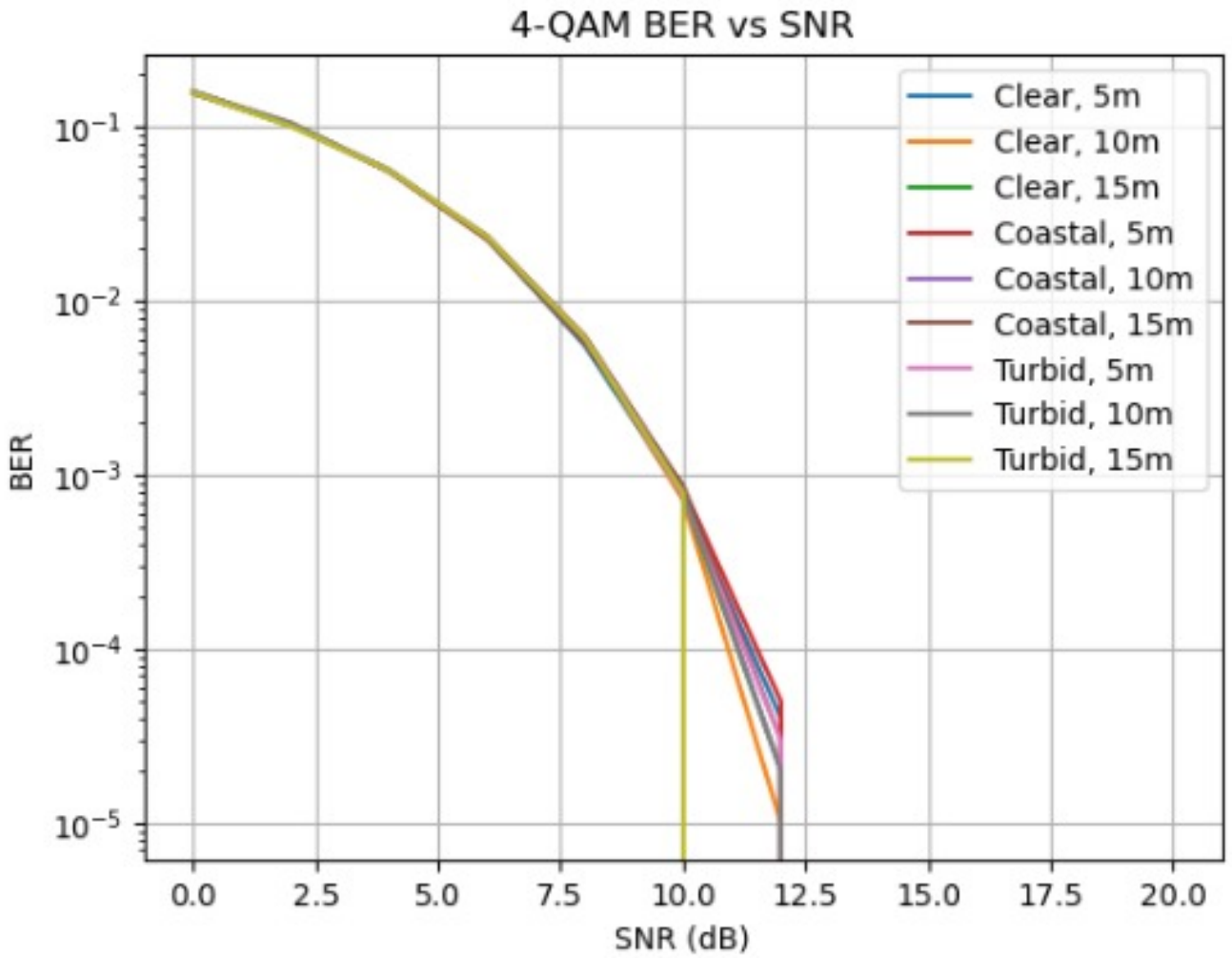
- BER decreases with SNR
- BER increases with distance
- Clear water performs best
- Turbid water performs worst
- Diversity improves performance

OOK Results



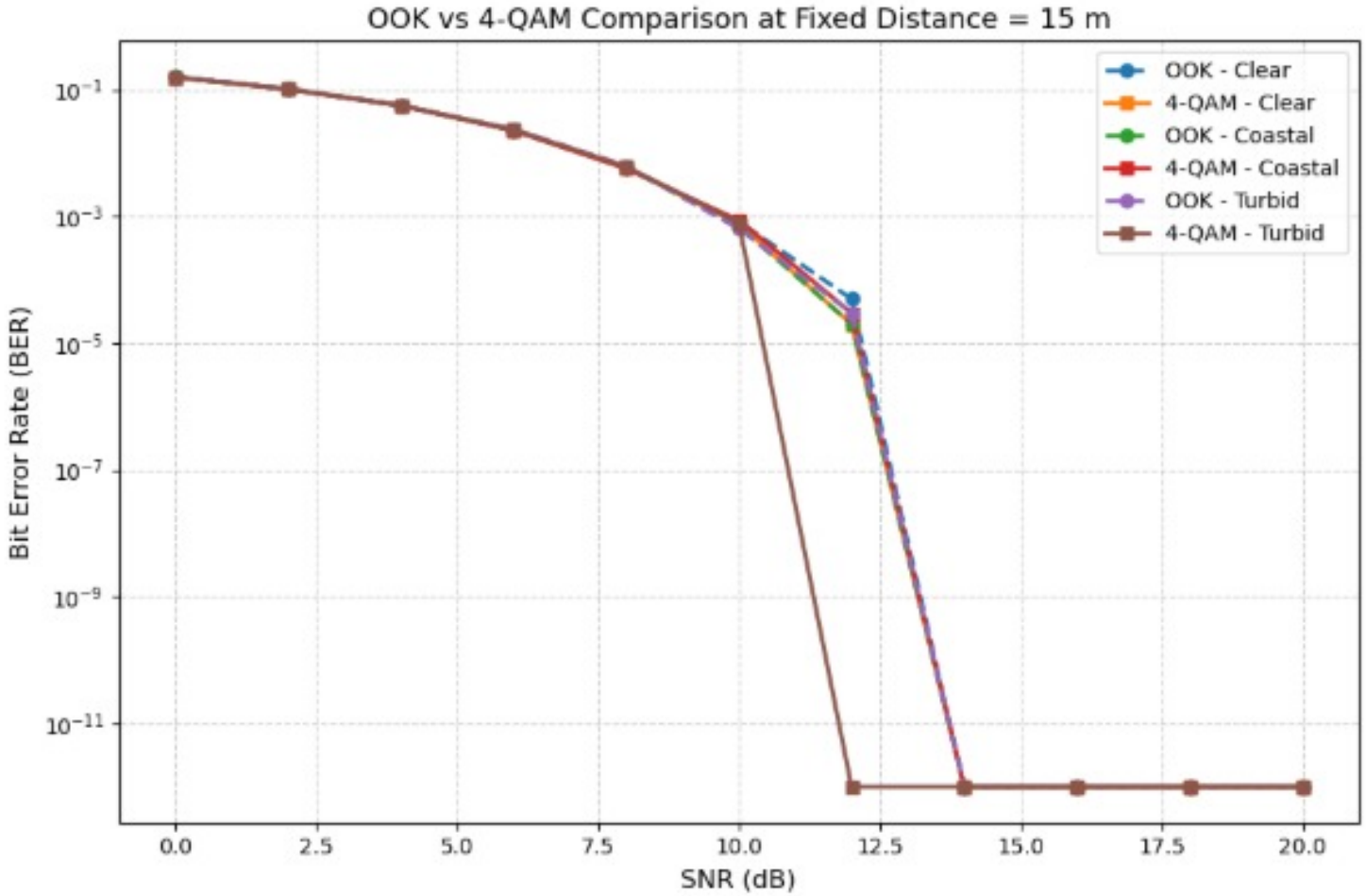
BER vs SNR for OOK

4-QAM Results



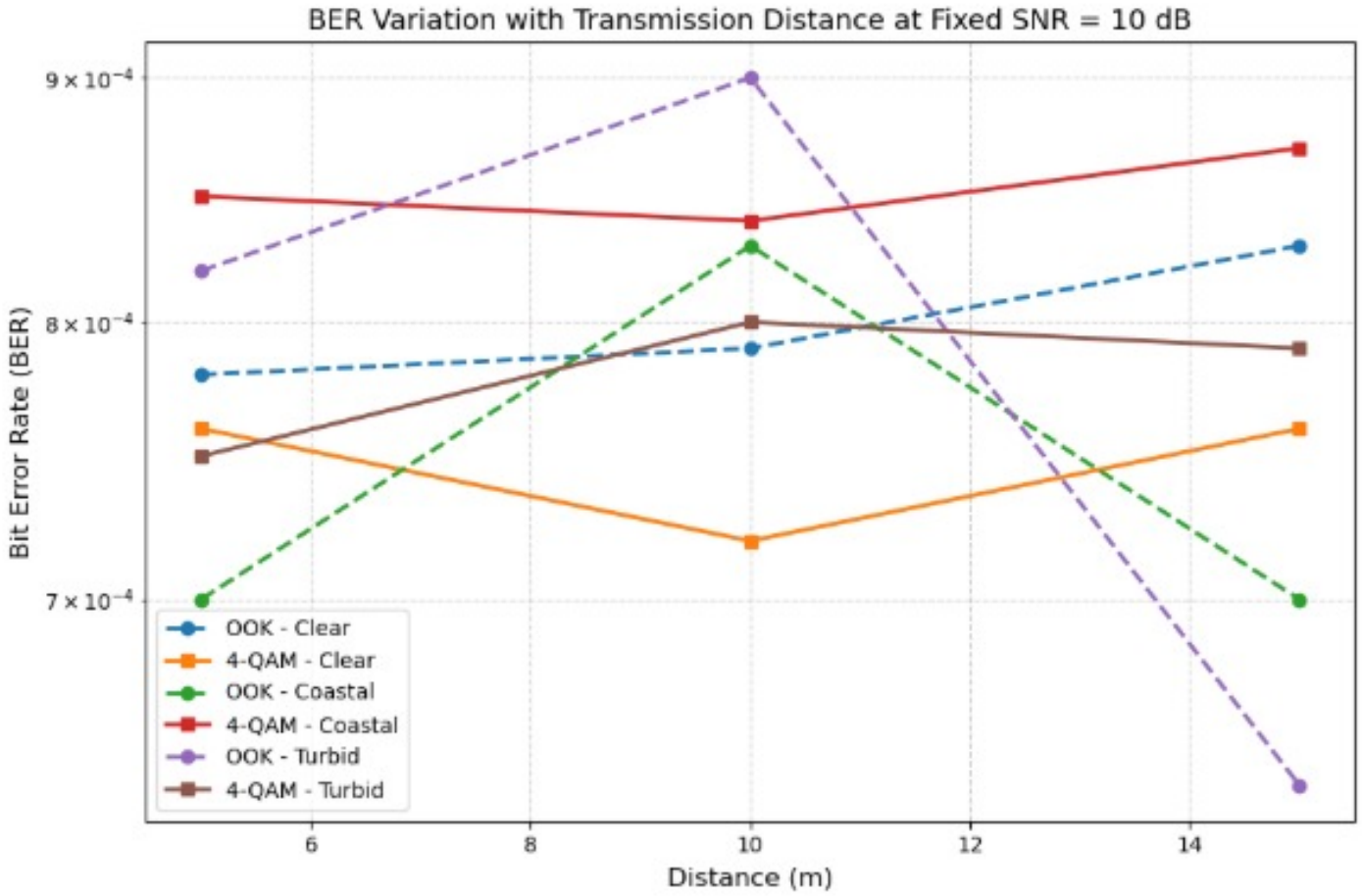
BER vs SNR for 4-QAM

Comparison



OOK vs 4-QAM

BER vs Distance (SNR = 10 dB)



BER variation with transmission distance at fixed SNR = 10 dB

Conclusion

- UWOC system analyzed using OOK and 4-QAM
- BER studied for SNR, distance, and water types
- Clear water gives best performance
- Diversity improves reliability

Future Work

- OFDM implementation
- MIMO extension
- Turbulence modeling
- ML-based detection

Thank You